

Ableitungen

- $(ax^b)' = abx^{b-1}$, $(a \ln x)' = \frac{a}{x}$,
 $(ae^x)' = ae^x$
- Produkt $(fg)' = f'g + fg'$; Quotient
 $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- Kette $(g(h(x)))' = h'(x)g'(h(x))$
- $(\sin)' = \cos$, $(\cos)' = -\sin$, $(\tan)' = \frac{1}{\cos^2 x} = 1 + \tan^2 x$
- $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, $(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$
- $(\arctan x)' = \frac{1}{1+x^2}$
- $(\sinh x)' = \cosh x$, $(\cosh x)' = \sinh x$,
 $(\tanh x)' = \frac{1}{\cosh^2 x}$
- $(A \sin \omega t)' = A\omega \cos \omega t$, $(A \cos \omega t)' = -A\omega \sin \omega t$

Vektoranalysis

Totales Differential / Gradient

$$df = \partial_x f dx + \partial_y f dy, \quad |\vec{v}| = \sqrt{v_1^2 + \dots + v_n^2}$$

$$\nabla f = (\partial_{x_1} f, \dots, \partial_{x_n} f)^\top$$

Richtungsableitung

$$D_{\vec{v}} f = \nabla f \cdot \frac{\vec{v}}{|\vec{v}|} \quad \blacktriangleright \nabla f \rightarrow \text{Punkt} \rightarrow \vec{v}$$

normieren $\rightarrow$ Skalarprodukt

Divergenz & Rotation

$$\operatorname{div} \vec{f} = \partial_x f_1 + \partial_y f_2 (+ \partial_z f_3)$$

$$\operatorname{rot}_{2D} \vec{f} = \partial_x f_2 - \partial_y f_1 \quad (\text{Skalar})$$

$$\operatorname{rot}_{3D} \vec{f} = \begin{pmatrix} \partial_y f_3 - \partial_z f_2 \\ \partial_z f_1 - \partial_x f_3 \\ \partial_x f_2 - \partial_y f_1 \end{pmatrix}$$

► Quellenfrei $\operatorname{div} = 0$; wirbelfrei $\operatorname{rot} = 0$;
Wurzel $\Rightarrow \pm$

Weitere

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)^\top$$

$$\Delta f = \operatorname{div}(\operatorname{grad} f) = \sum_i \partial_{x_i}^2 f$$

$$\operatorname{rot}(\operatorname{grad} f) = \vec{0}, \quad \operatorname{div}(\operatorname{rot} \vec{f}) = 0$$

Extremstelle: $\nabla f = \vec{0}$ ► System mit Einsetzen lösen

Differentialgleichungen

Linear homogen $y' = a(x)y$

$$y = ce^{A(x)}, \quad A = \int a dx. \quad \text{Zerfall } y' = \alpha y \Rightarrow y_0 e^{\alpha t}, \quad t_{1/2} = \frac{\ln 2}{-\alpha}$$

Linear inhomogen $y' = ay + b$

$$y = e^A(y_0 + \int_{x_0}^x b(u)e^{-A(u)} du)$$

► $a, b \rightarrow A = \int a \rightarrow e^A, e^{-A}$ kürzen $\rightarrow$ Integral m. Grenzen $\rightarrow$ einsetzen

Trennung $y' = g(x)h(y)$

$$\int \frac{1}{h(y)} dy = \int g(x) dx \quad \blacktriangleright \text{nach } y \text{ auflösen}$$

$\rightarrow c$ aus Startwert

Integralrechnung

$$\int_a^b f dx = F(b) - F(a)$$

$$\bullet \int ax^b dx = \frac{a}{b+1} x^{b+1} \quad (b \neq -1), \quad \int \frac{1}{x} dx = \ln|x|$$

$$\bullet \int e^x dx = e^x, \quad \int f' e^f dx = e^f, \\ \int \ln x dx = x \ln x - x$$

$$\bullet \int \sin x dx = -\cos x, \quad \int \cos x dx = \sin x$$

Partielle Integration

$$\int uv' dx = uv - \int u'v dx \quad \blacktriangleright u = \text{Potenz}/\ln, \\ v' = \text{Rest}; \text{Produkt bleibt} \rightarrow \text{nochmal}$$

Substitution

$$\int f(g)g' dx = \int f(u)du \quad \blacktriangleright u = \text{innere Fkt.}, \quad du = u' dx, \quad \text{kürzen, integrieren, zurück}$$

Doppelintegral

$$\int_{a_y}^{b_y} \int_{a_x}^{b_x} f dx dy \quad \blacktriangleright \text{innen nach } x \text{ (y konst.,} \\ \int c dx = cx), \text{ dann nach } y$$

Trigonometrie

$$\sin t, \cos t : \mathbb{R} \rightarrow [-1, 1], \quad \text{Periode } 2\pi,$$

$$\cos t = \sin(t + \frac{\pi}{2})$$

$$\sin^2 t + \cos^2 t = 1, \quad \tan t = \frac{\sin t}{\cos t}, \quad 1 + \tan^2 t = \frac{1}{\cos^2 t}$$

$$\cosh^2 t - \sinh^2 t = 1, \quad \tanh t = \frac{\sinh t}{\cosh t}$$

$$\sec = \frac{1}{\cos}, \quad \csc = \frac{1}{\sin}, \quad \cot = \frac{1}{\tan}$$

$$\text{Welle: } \lambda = \frac{1}{f} = \frac{2\pi}{\omega}, \quad \omega = 2\pi f$$

$$\sin: 0, 1, 0 \quad \cos: 1, 0, -1 \quad (0, \frac{\pi}{2}, \pi)$$

► Trig-Integral: $\tan \rightarrow \frac{\sin}{\cos}$, ausklammern,
 $\sin^2 + \cos^2 = 1$, integrieren

Komplexe Zahlen

$$i^2 = -1, \quad z = a + ib, \quad \operatorname{Re} z = a, \quad \operatorname{Im} z = b,$$

$$\bar{z} = a - ib$$

$$|z| = \sqrt{a^2 + b^2} \quad (\text{beide quadrieren!})$$

$$z_1 z_2 = (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + a_2 b_1)$$

$$z^2 = (a^2 - b^2) + 2abi$$

$$\frac{z_1}{z_2} = \frac{(a_1 a_2 + b_1 b_2) + i(a_2 b_1 - a_1 b_2)}{a_2^2 + b_2^2}$$

► mit $\bar{z}$ erweitern

$$z = r(\cos \varphi + i \sin \varphi) = re^{i\varphi}, \quad r = |z|, \\ \varphi = \arctan \frac{b}{a}$$

$$e^{x+iy} = e^x(\cos y + i \sin y), \quad z^x = r^x e^{i\varphi x}, \\ e^{i\pi} + 1 = 0$$

Fourier

$$A(\cos \omega t + i \sin \omega t) = Ae^{i\omega t}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$